Asem Sharif

AI | Computer Vision Junior Engineer

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Profile

Junior **Artificial Intelligence and Computer Vision** student with practical experience in **image and video analysis systems**. Skilled in data-driven problem solving, algorithm development, and performance evaluation, with a focus on building reliable and accurate solutions.

Additionally experienced in **Desktop Application Development**, creating structured and maintainable software with attention to usability and clean code.

Education

Faculty of Artificial Intelligence – Menoufia University Oct 2022 – Aug 2026

- **Bachelor of Science** in Artificial Intelligence - Cyber Security Department. 120 Hours
- **GPA**: **3.55** / 4.0 (**A-Excellent** Standing).
- **Academic Coursework**:
  - Background in mathematics, programming, and software engineering, in addition to advanced studies in artificial intelligence, machine learning, deep learning, computer vision, and data analysis, along with specialization in cyber security, including cryptography, secure software development, and network security.

Technical Skills

Programming — Python, C++, GO, Object-Oriented Programming, Algorithm Design and Analysis

Artificial Intelligence — Machine Learning, Deep Learning, Neural Networks, Transformers, Automation

Computer Vision — Image Processing, CNNs, Vision Transformer (ViT), Object Detection, Tracking, Segmentation

Application Development — Desktop Application Development, Event-Driven Architecture, Software Maintenance

Frameworks — PyTorch, TensorFlow, Keras, MediaPipe, Dlib, YOLO, OpenCV, Scikit-Learn, Scikit- Image | PyQt

Edge Deployment — Raspberry Pi 5, Hailo-8L NPU, Edge AI Inference, Quantization, Pruning, Hardware Prototyping

Work Experience

Junior AI Developer | Research Assistant 2024 – Present

- Contributed to end-to-end machine learning workflows, including model evaluation using appropriate performance metrics, iterative tuning on real-world datasets to improve accuracy and generalization, and systematic hyperparameter optimization to enhance model robustness and reliability. Freelance

Instructor at Digital Egypt Cubs Initiative 2025

- Delivered interactive sessions on foundational computer science and digital technology concepts, simplifying topics such as hardware, software, and internet safety. Developed and integrated small educational to reinforce learning and inter-connect concepts. DECI
- Contributed to the national initiative **DECI** 🔗, powered by **Udacity** 🔗 and **CLS** 🔗, aimed at enhancing digital literacy among youth.

University Projects

Undergraduate 2022 – 2025

A collection of academic assignments and projects covering programming, artificial intelligence, and desktop applications development.

Graduation Project 2026

An end-to-end computer vision system for real-time driver behavior analysis and vehicle safety. (Currently Private)

- **Lynx DMS - Intelligent Edge Deployed Driver Monitoring System**
  - Real-time computer vision system for driver state, behavior, and safety monitoring.
  - Implements:
    - Face recognition for authorized access control and driver validation.
    - Safety and emotion analysis modules to assess driver condition in real-time.
    - Continuous multiple class detection pipeline with event-driven alert system.
    - Dashboard for local processing and logging to ensure privacy preserving serving.

  Achievement: Delivered production-ready optimized edge prototype capable of sub-30ms latency and over 95% accuracy with low false positives for critical safety alerts.

  Interoperability: Engineered a structured event-flagging engine that generates real-time descriptions and triggers; designed for seamless integration with external hardware controllers, contact interfaces, or fleet management systems.

  Technical Stack:
  Python, TensorFlow, PyTorch, MediaPipe, OpenCV, Scikit-Image
  PyQt6, SQLite, ONNX, Raspberry Pi 5, Hailo-8L NPU.

  Source Code: github.com/asem-sharif-ai/Lynx-DMS